



Network Layer : Addressing

Dr. Alekha Kumar Mishra



Network Layer

- Concerned with getting packets from the **source** all the way to the **destination**
- The packet may pass through several LANs or WANs before reaching the destination computer
- The network layer must know about the **topology** of the network and choose appropriate paths through it, even for large networks
- Choosing a **route** is crucial
- Need a **global addressing scheme**



Logical addressing

- In TCP/IP protocol suite, **IP address** means a logical address in the network layer
- An IPv4 address **uniquely** and **universally** defines the **connection** of a device (a computer or a router) to the Internet
- Each address defines one, and only one, connection to the Internet
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296
- Binary notation (4-byte address)
 - 01110101 10010101 00011101 00000010
- Dotted-decimal notation.
 - 117.149.29.2



Classful Addressing

- This scheme is becoming obsolete
- The address space is divided into five classes: A, B, C, D, and E. Each class occupies some part of the address space

	First byte	Second byte	Third byte	Fourth byte
Class A	0			
Class B	10			
Class C	110			
Class D	1110			
Class E	1111			

a. Binary notation

	First byte	Second byte	Third byte	Fourth byte
Class A	0–127			
Class B	128–191			
Class C	192–223			
Class D	224–239			
Class E	240–255			

b. Dotted-decimal notation

Number of blocks and block size in classful addressing



<i>Class</i>	<i>Number of Blocks</i>	<i>Block Size</i>	<i>Application</i>
A	128	16,777,216	Unicast
B	16,384	65,536	Unicast
C	2,097,152	256	Unicast
D	1	268,435,456	Multicast
E	1	268,435,456	Reserved



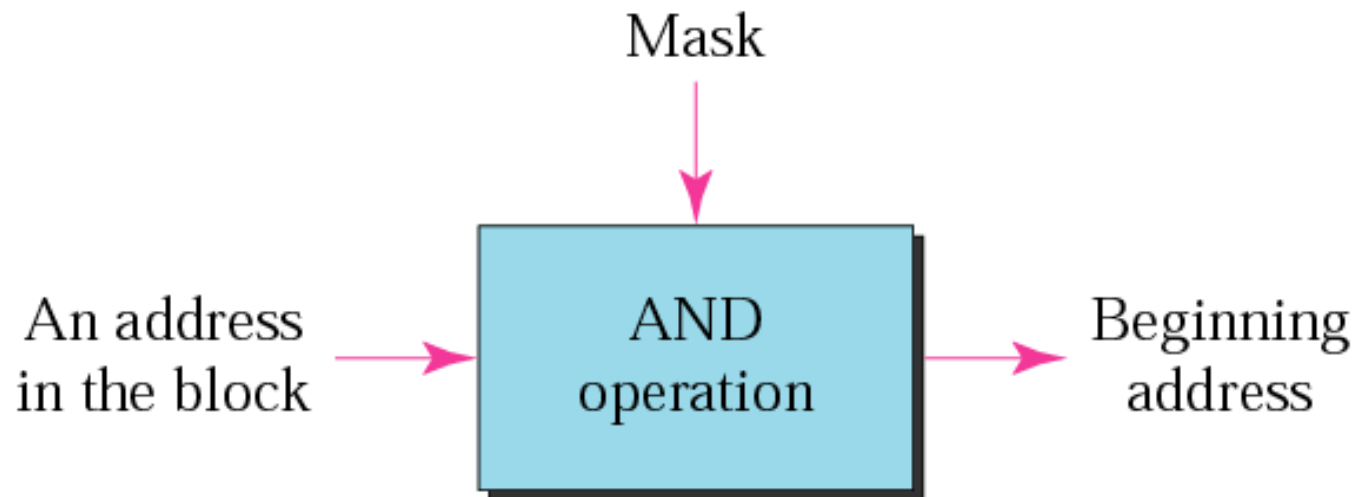
Network Address

- The **first address** of a block of addresses is called the network address and defines the organization network
- The network address **defines** the network to the **rest of the Internet**.
- Given the network address, we can find the **class** of the address, the **block**, and the **range** of the addresses in the block
- The first address is the one that is used by routers to direct the message sent to the organization from the outside



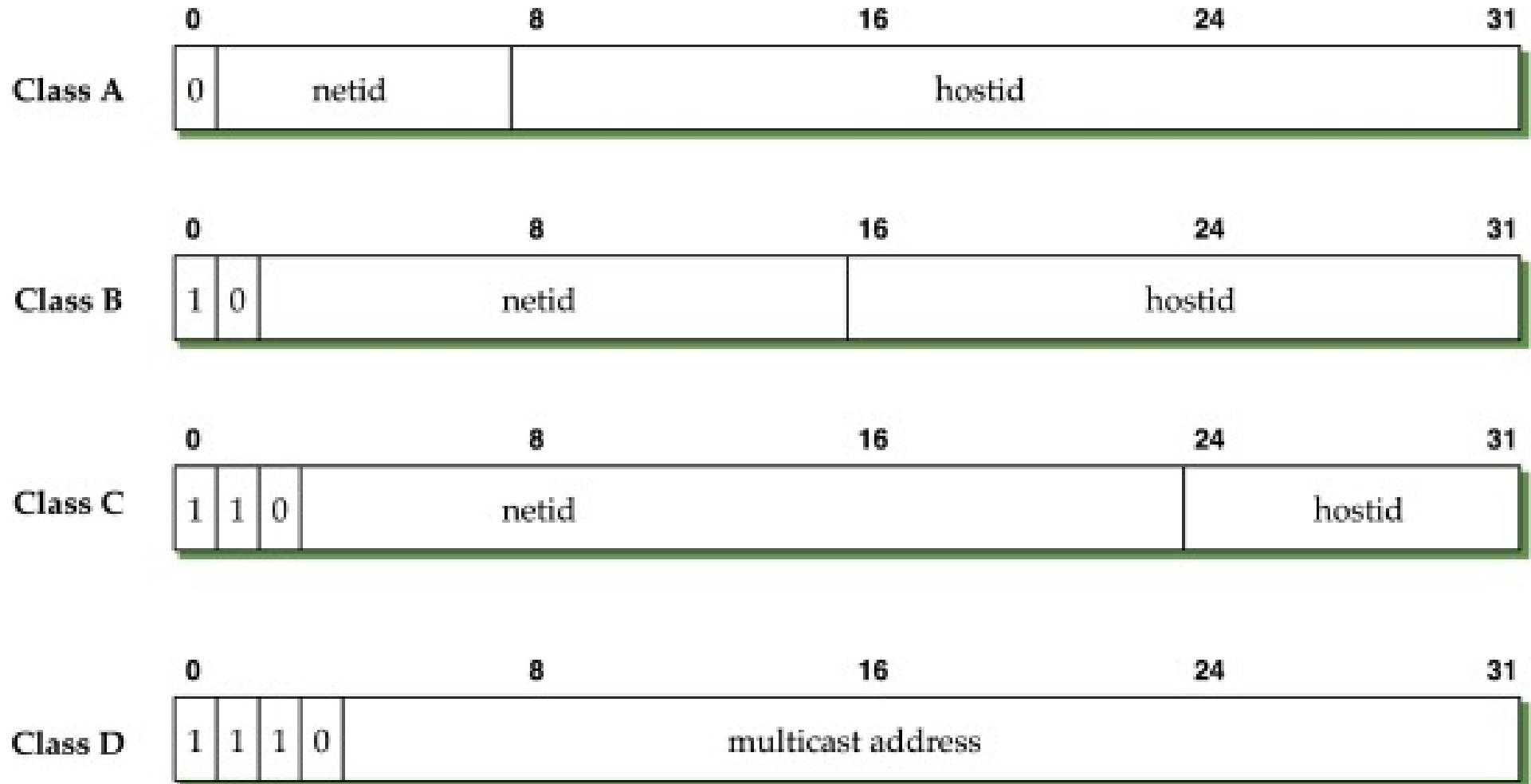
Netid and Hostid

- An IP address in class A, B, or C is divided into **netid** and **hostid**.
- These parts are of varying lengths, depending on the class of the address
- The concept does not apply to classes D and E
- We can also use a **mask** (also called the **default mask**), a 32-bit number made of contiguous 1s followed by contiguous 0s
- A mask is a 32-bit number in which the n leftmost bits are 1s and the $32 - n$ rightmost bits are 0s.



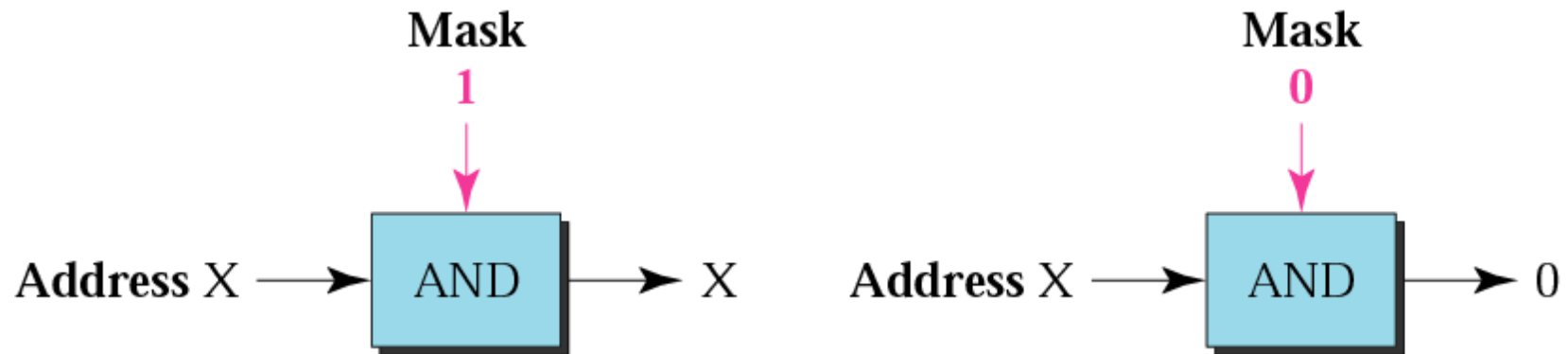


Netid and Hostid





Default mask



Class	Binary	Dotted-Decimal	CIDR
A	11111111 00000000 00000000 00000000	255.0.0.0	/8
B	11111111 11111111 00000000 00000000	255.255.0.0	/16
C	11111111 11111111 11111111 00000000	255.255.255.0	/24



- Given the address 23.56.7.91, find the beginning address (network address)
- Given the address 132.6.17.85 and, find the beginning address (network address)
- Given the address 201.180.56.5, find the beginning address (network address)



Loop back address

- IP address with first byte as 127[127.X.Y.Z]
- Packet sent to this address is not put on to the wire, it is processed locally as incoming packet
- Used to test the s/w on a machine [Here the packet simply returns to the protocol s/w on the same machine]
 - Eg. Ping to test if the process is able to receive and process a packet.
 - Eg. Client process sending message to server process on the same machine



Special Addresses

Special Address	Netid	Hostid	Source or Destination
Network address	Specific	All 0s	None
Direct broadcast address (broadcast on distant n/w)	Specific	All 1s	Destination
Limited broadcast address	All 1s	All 1s	Destination
The host on this network	All 0s	All 0s	Source
Specific host on this network	All 0s	Specific	Destination
Loopback address	127	Any	Destination



Subnetting

- A large block of addresses is splitted into several contiguous groups and assign each group to smaller networks for internal purpose called **subnets**
- **The process is called subnetting**
- Subnet still act like a single network to the outside world
- The subnet ID is generally used to group hosts based on the physical network topology
-
- **Subnetting** increases the number of 1s in the mask

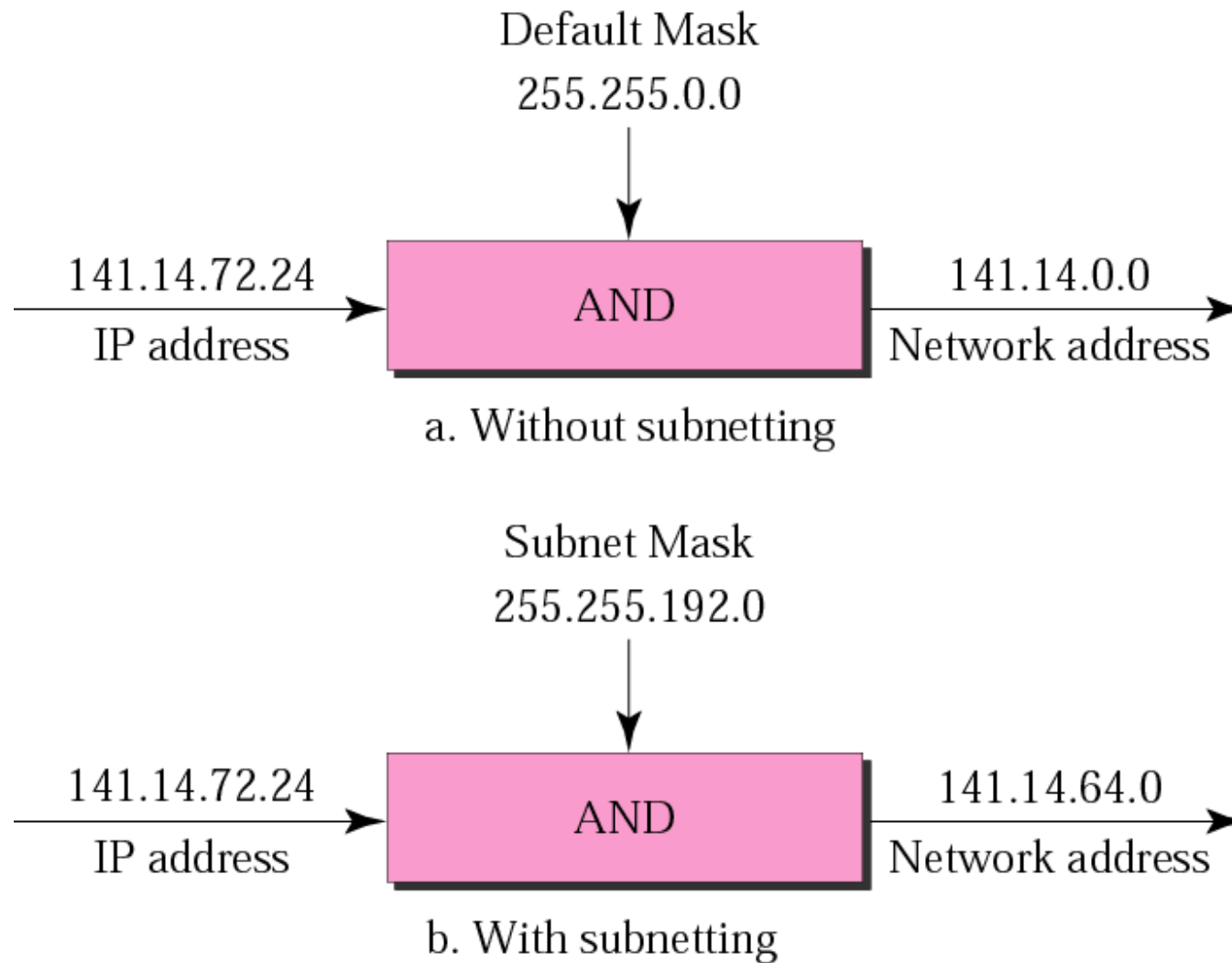


Subnetting

- By default, IPv4 address provides 2 level hierarchy: Net id and Host id
- To achieve 3 level hierarchy, subnetting is required
- The number of subnets must be a power of 2.

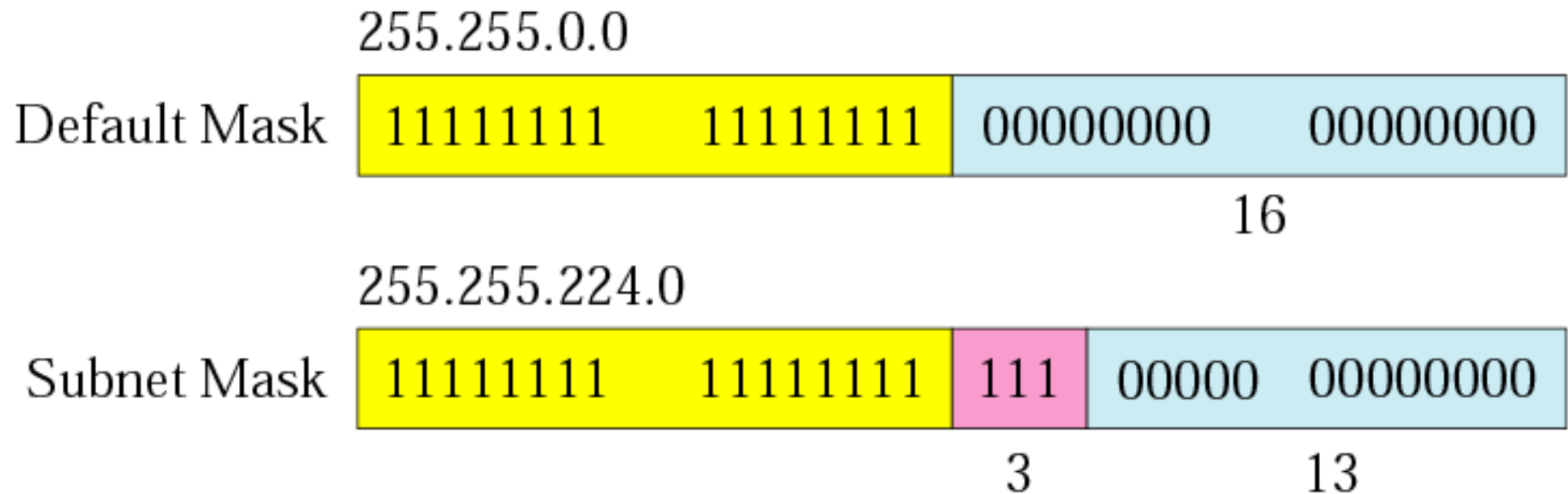


Default Mask vs Subnet Mask





Default vs Subnet Mask





Finding subnet address

- To find subnet address use binary notation for both the address and the mask and then apply the AND operation to find the subnet address.
- What is the subnetwork address if the destination address is 200.45.34.56 and the subnet mask is 255.255.240.0?

Alternative way

- If the byte in the mask is 255, copy the byte in the address
- If the byte in the mask is 0, replace the byte in the address with 0.
- If the byte in the mask is neither 255 nor 0, we write the mask and the address in binary and apply the AND operation.

IP Address									
19	•	30	•	84	•	5			
Mask									
255	•	255	•	192	•	0			
19	•	30	•	64	•	0			
Subnet Address									

84	0	1	0	1	0	1	0	0
192	1	1	0	0	0	0	0	0
64	0	1	0	0	0	0	0	0



Example 1

- A company is granted the site address 201.70.64.0 (class C). The company needs six subnets. Design the subnets.
- The company needs six subnets
- 6 is not a power of 2
 - The next number that is a power of 2 is 8 (2^3)
 - We need 3 more bits to identify subnets
 - Hence, we add 3 additional bit in the subnet mask from the leftmost zeros.
- The total number of 1s in the subnet mask is 27 ($24+3$).
- The total number of 0s in subnet mask is 5 ($32-27$).

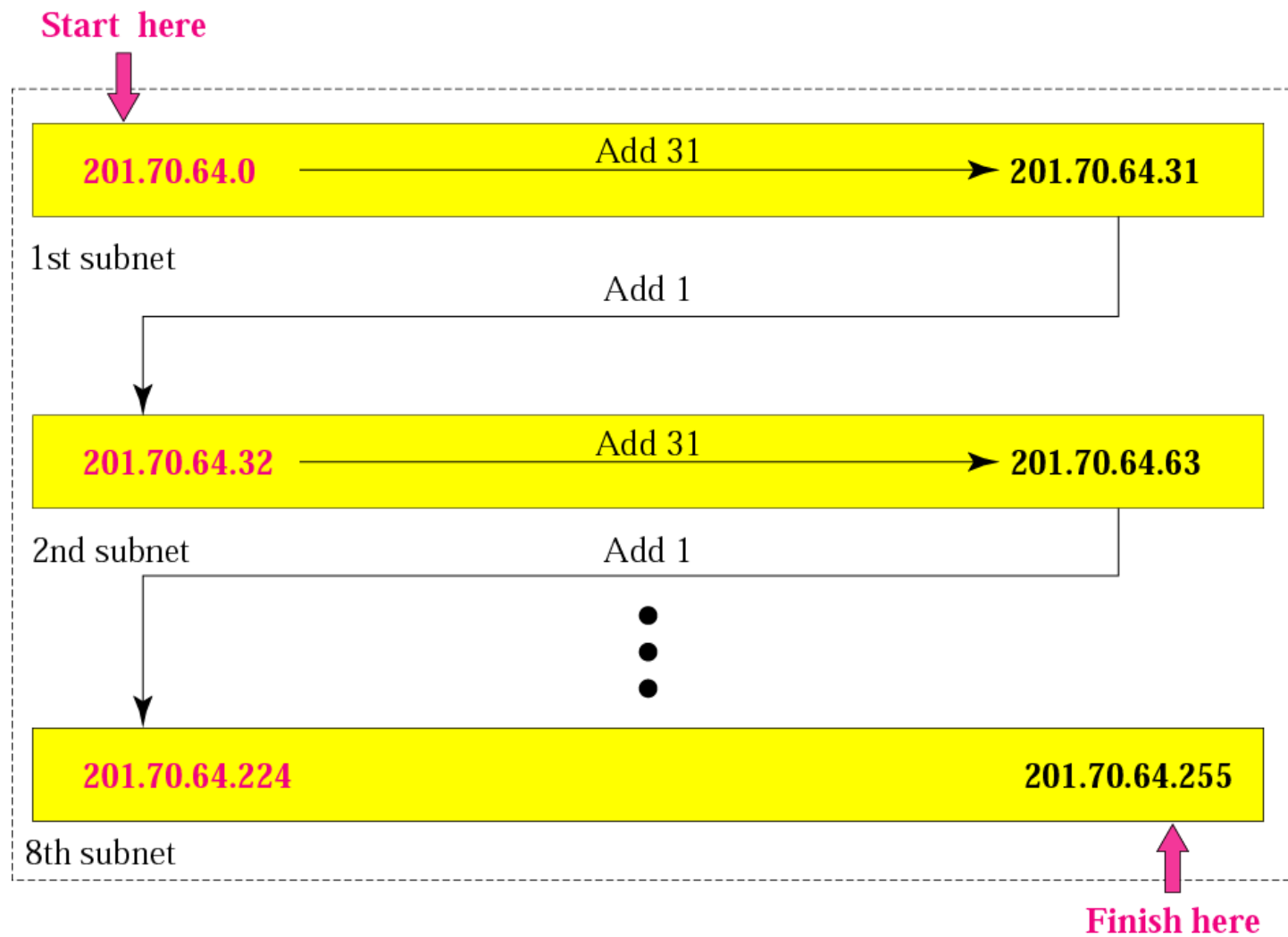


Example 1

- The new mask is
 - 11111111 11111111 11111111 11100000
 - or
 - 255.255.255.224
- The number of subnets is 8.
- The number of addresses (hosts) in each subnet is 2^5 or 32.



Example 1





Example 2

- A company is granted the site address 181.56.0.0 (class B). The company needs 1000 subnets. Design the subnets

-
-
-

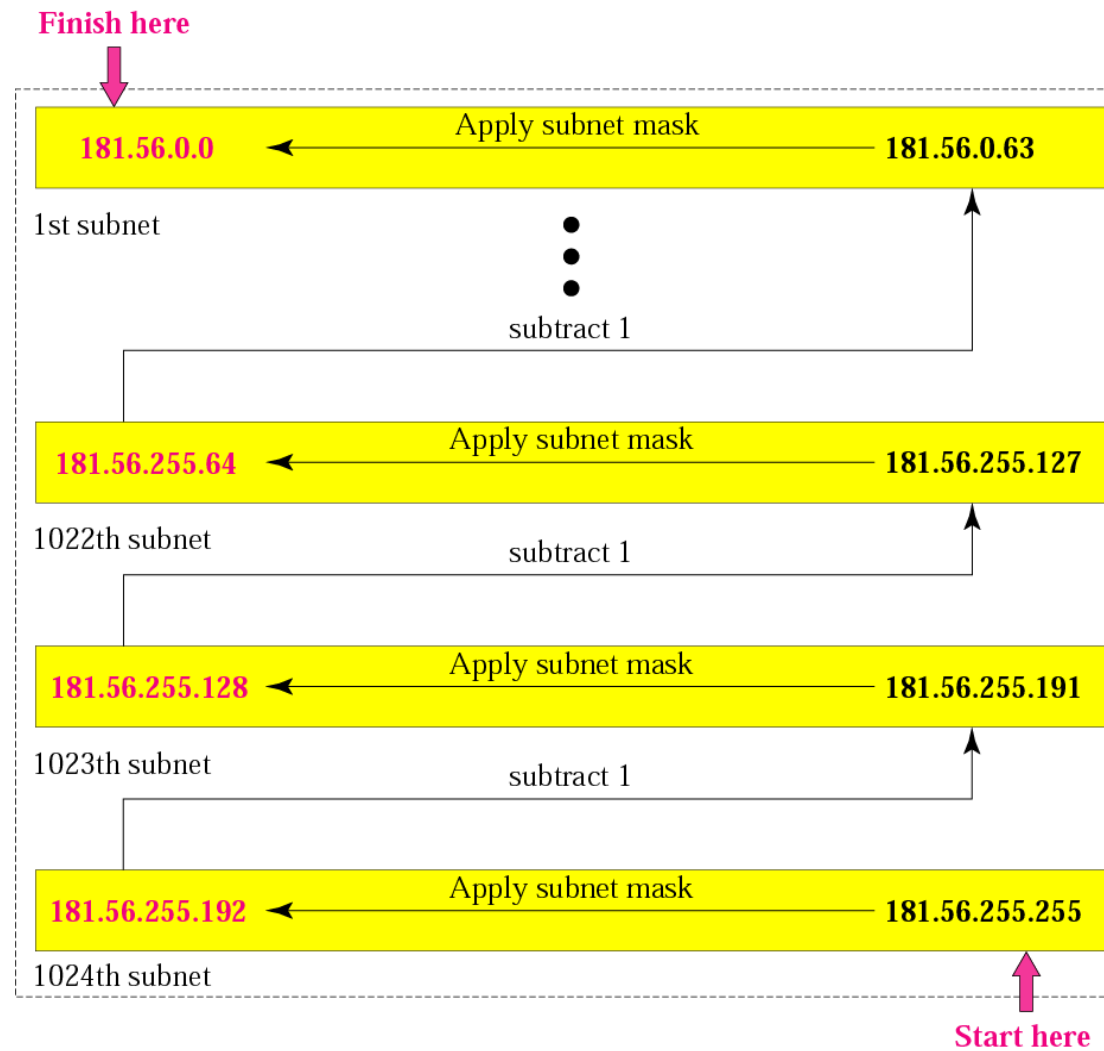


Example 2

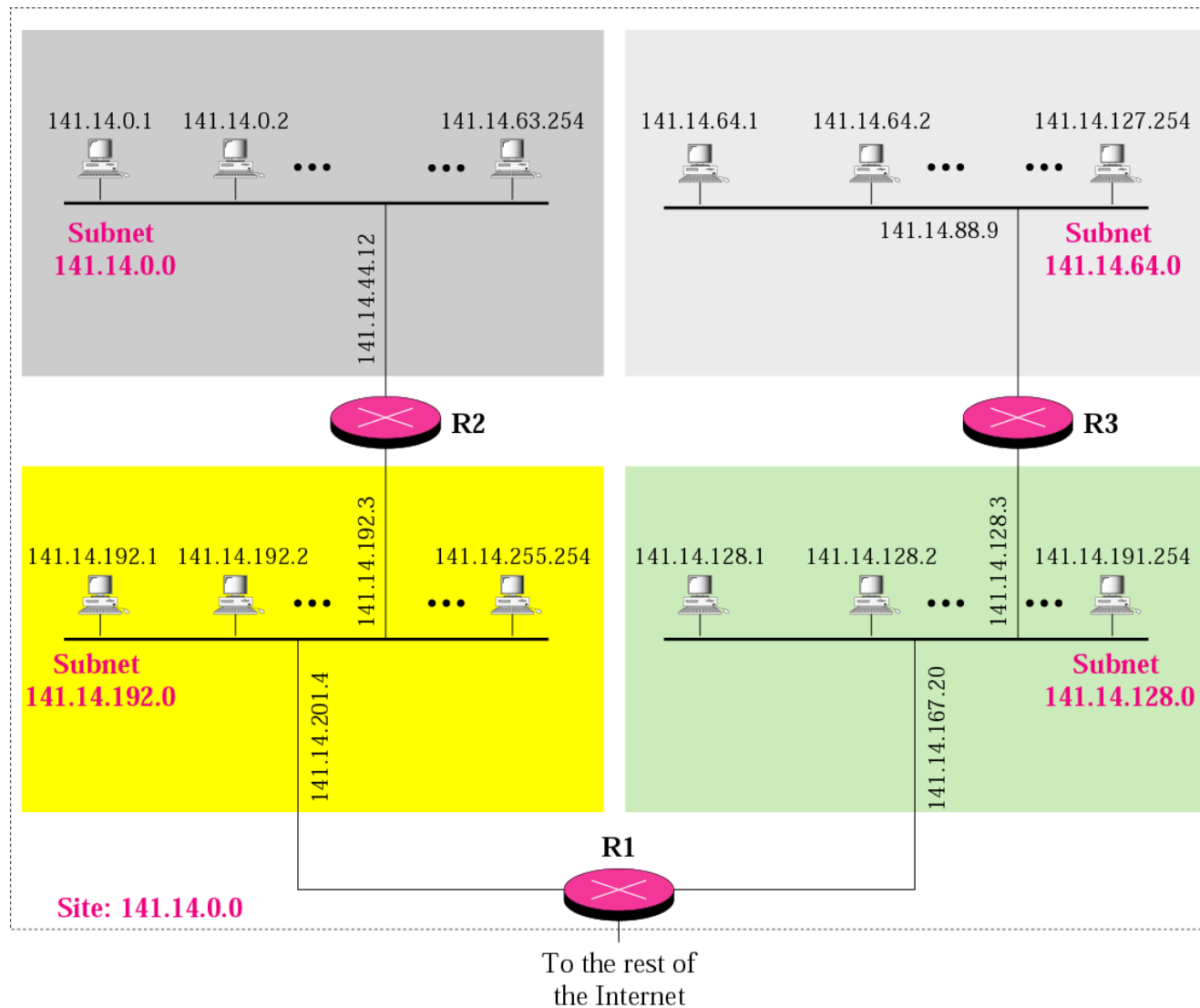
- The company needs 1000 subnets
 - The next number that is a power of 2 is 1024 (2^{10}). We need 10 more 1s in the subnet mask
- The total number of 1s in the subnet mask is 26 ($16+10$).
- The total number of 0s is 6 ($32-26$)
- The mask is
- **11111111 11111111 11111111 11000000**
- or
- **255.255.255.192.**
- The number of subnets is 1024
- The number of addresses in each subnet is 2^6 or 64.



Example 2



A network with three levels of hierarchy (subnetted)





supernetting

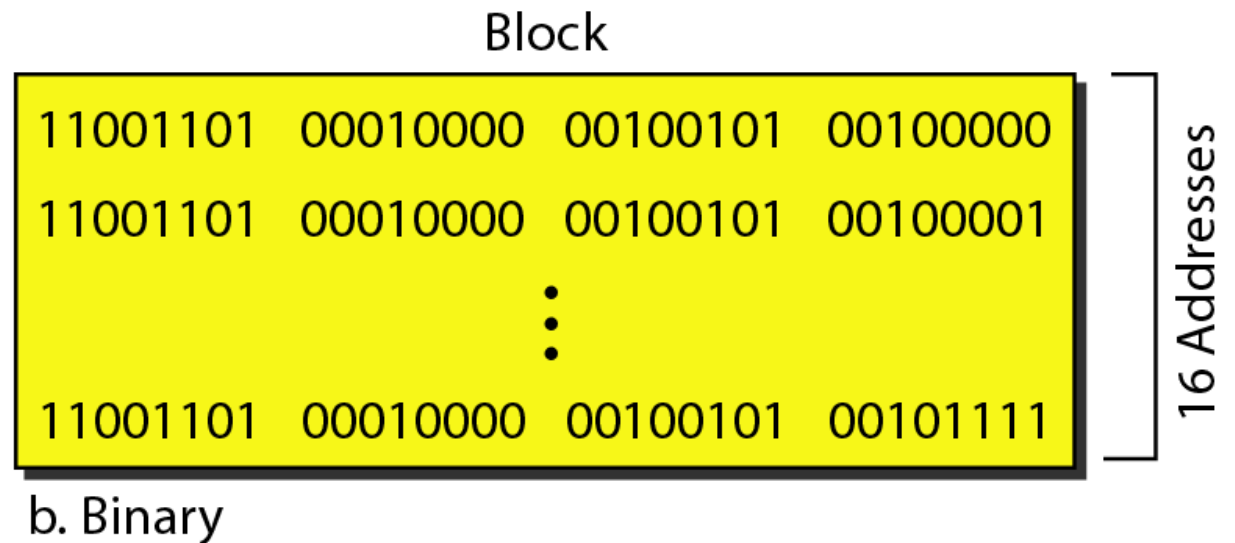
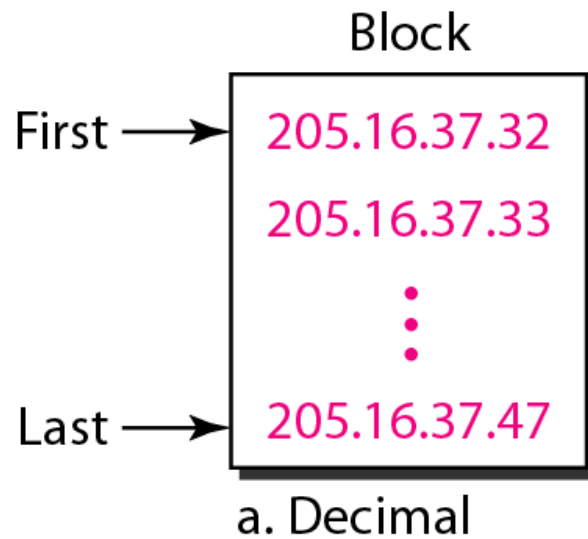
- The time came when most of the class A and class B addresses were depleted
- However, there was still a huge demand for midsize blocks (class C)
- In **supernetting**, an organization can combine several class C blocks to create a larger range of addresses
- Supernetting decreases the number of 1s in the mask
 - For example, an organization that needs 1000 addresses can be granted four contiguous class C blocks. The mask changes from /24 to /22
- **Classless addressing** eliminates the need for supernetting



Classless Addressing

- Flaws in classful addressing led to address depletion
- One solution to the problem is the idea of **classless addressing**
- There are no classes, but the addresses are still granted in blocks.
- When an entity, small or large, needs to be connected to the Internet, it is granted a block (range) of addresses
- The size of the block (the number of addresses) varies based on the nature and size of the entity
- Restriction
 - The addresses in a block must be contiguous, one after another.
 - The number of addresses in a block must be a power of 2 (1, 2, 4, 8, ...).
 - The first address must be evenly divisible by the number of addresses

Classless Addressing





Mask in Classless Addressing

- In classless addressing the mask for a block can take any value from 0 to 32
- It is very convenient to give just the value of n preceded by a slash called Classless Inter-Domain Routing (**CIDR**) notation
 - **$x.y.z.t/n$**
 - **Prefix** -> The common part of the address range(similar to *netid*)
 - **Prefix length** -> length of the prefix $/n$
 - **Suffix** -> varying part(similar to *hostid*)
 - **Suffix length** -> length of the suffix $(32-n)$



Prefixes

<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>	<i>/n</i>	<i>Mask</i>
/1	128.0.0.0	/9	255.128.0.0	/17	255.255.128.0	/25	255.255.255.128
/2	192.0.0.0	/10	255.192.0.0	/18	255.255.192.0	/26	255.255.255.192
/3	224.0.0.0	/11	255.224.0.0	/19	255.255.224.0	/27	255.255.255.224
/4	240.0.0.0	/12	255.240.0.0	/20	255.255.240.0	/28	255.255.255.240
/5	248.0.0.0	/13	255.248.0.0	/21	255.255.248.0	/29	255.255.255.248
/6	252.0.0.0	/14	255.252.0.0	/22	255.255.252.0	/30	255.255.255.252
/7	254.0.0.0	/15	255.254.0.0	/23	255.255.254.0	/31	255.255.255.254
/8	255.0.0.0	/16	255.255.0.0	/24	255.255.255.0	/32	255.255.255.255



Ways to find the first and the last address

- The address and the /n notation completely define the whole block
- **Decimal Notation**
 - The first address in the block can be found by setting the 32 - n right-most bits in the binary notation of the address to 0s.
 - The last address in the block can be found by setting the 32 - n right-most bits in the binary notation of the address to 1s
 - The number of addresses in the block can be found by using the formula 2^{32-n}
- **Binary Notation**
 - The first address can be found by **ANDing** the given addresses with the mask
 - The last address can be found by **ORing** the given addresses with the complement of the mask
 - The number of addresses can be found by **complementing** the mask, interpreting it as a decimal number, and **adding 1** to it



Example 3

- Find the first, last address, and number of addresses of the block 205.16.37.39/28
- The binary representation of the given address is 11001101 00010000 00100101 00100111
- By setting 32–28 rightmost bits to 0, we get
 - 11001101 00010000 00100101 00100000
 - 205.16.37.32
- By setting 32 – 28 rightmost bits to 1, we get
 - 11001101 00010000 00100101 00101111
 - 205.16.37.47
- The value of n is 28, hence number of addresses is 2^{32-28} or 16
- Repeat the above example using logical operations on binary notation of the above address



exercise

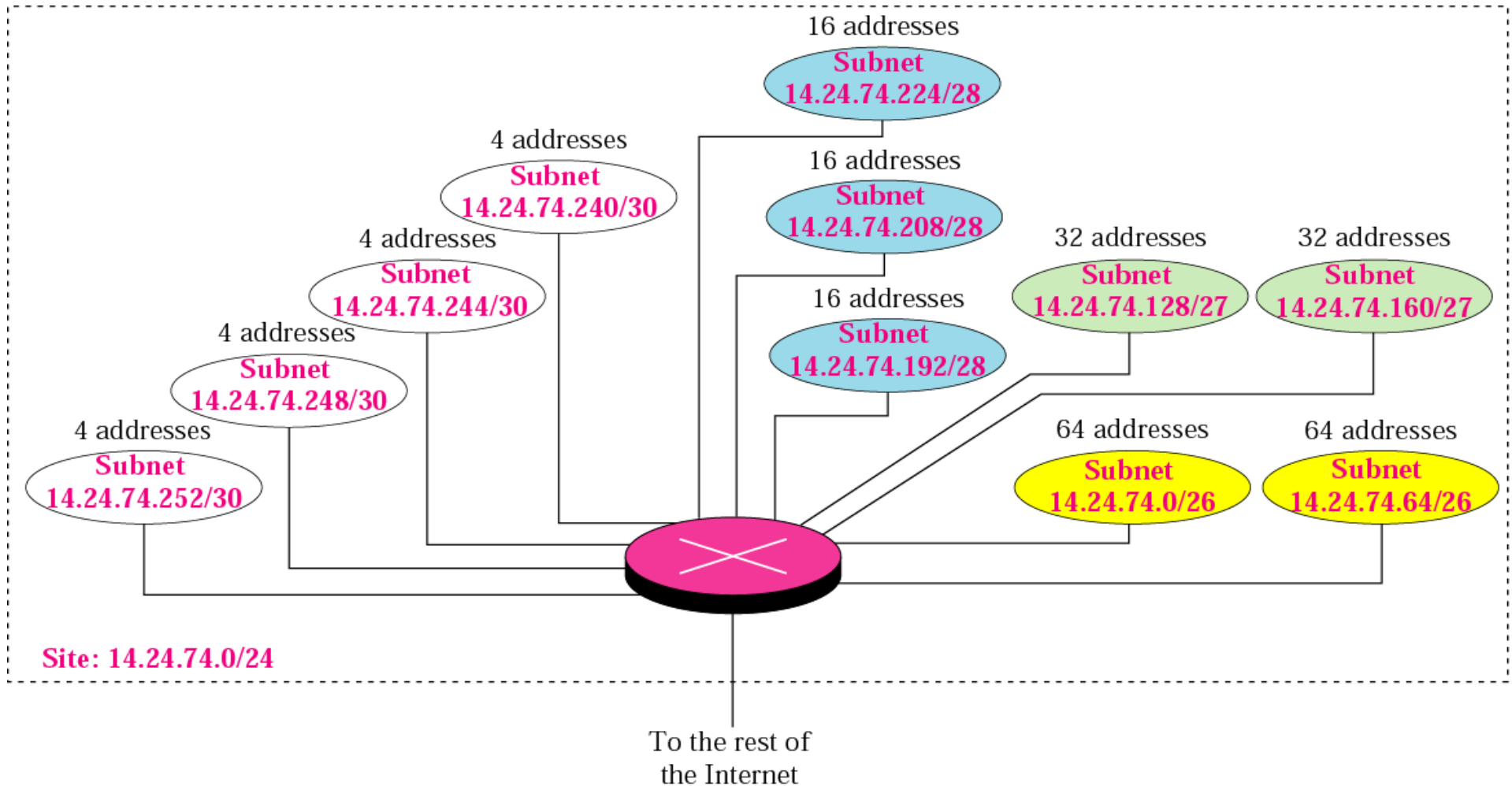
- Find the first and last and number of addresses in the block if one of the addresses is 140.120.84.24/20.



Example 4

- An organization is granted a block of addresses with the beginning address 14.24.74.0/24. There are $2^{32-24} = 256$ addresses in this block. The organization needs to have 11 subnets as shown below:
 - two subnets, each with 64 addresses.
 - two subnets, each with 32 addresses.
 - three subnets, each with 16 addresses.
 - four subnets, each with 4 addresses.
- Design the subnets.

Example 4

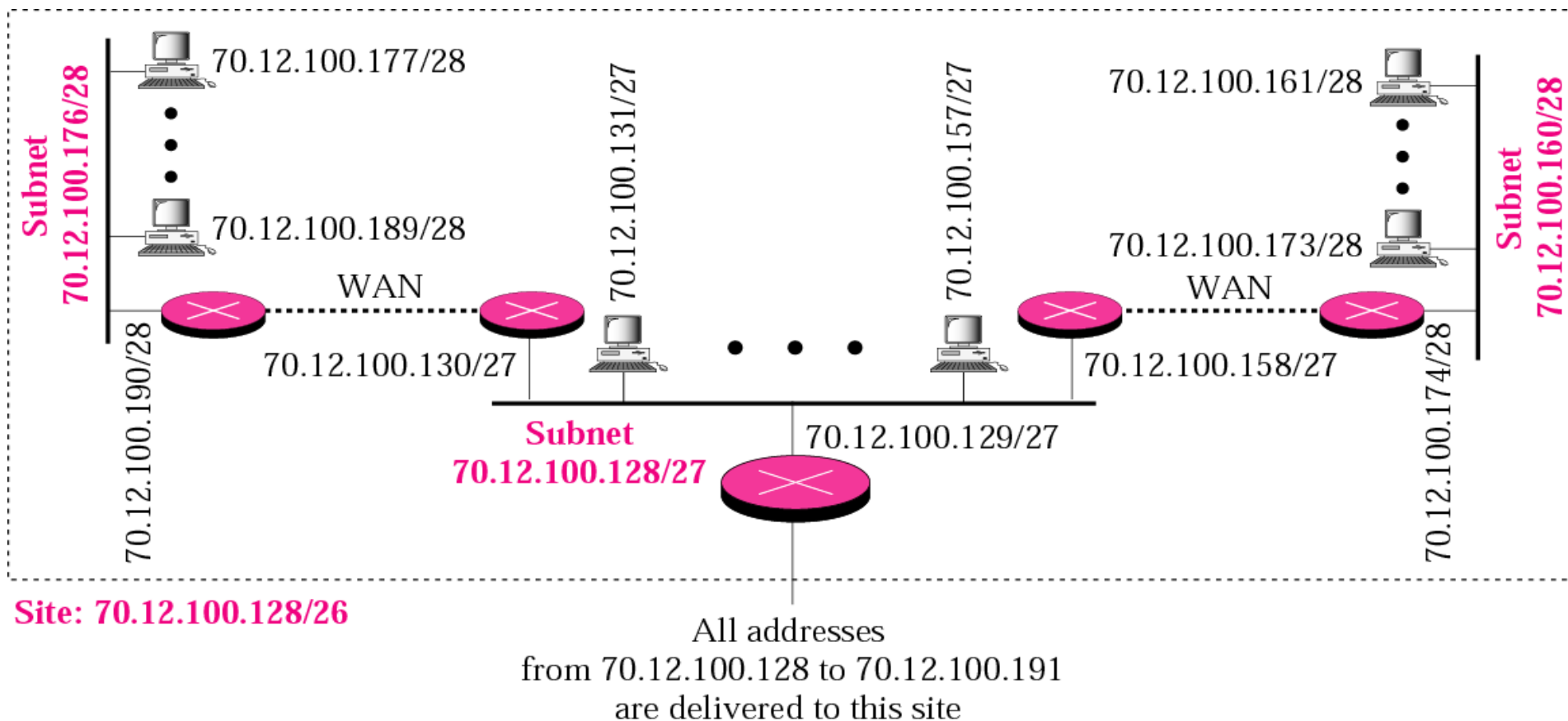




Example 5

- Assume a company has three offices: Central, East, and West. The Central office is connected to the East and West offices via private, point-to-point WAN lines. The company is granted a block of 64 addresses with the beginning address 70.12.100.128/26. The management has decided to allocate 32 addresses for the Central office and divides the rest of addresses between the two offices.

Example 5





Address allocation by ISP

- Address allocation is the responsibility of a global authority called the **Internet Corporation for Assigned Names and Addresses (ICANN)**. It usually assigns a large block of addresses to an ISP to be distributed to its Internet users.
- Many blocks of addresses are aggregated in one block and granted to one ISP called **Address aggregation**



Example 6

- An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:
- The first group has 64 customers; each needs 256 addresses.
- The second group has 128 customers; each needs 128 addresses.
- The third group has 128 customers; each needs 64 addresses.
- Design the subblocks and find out how many addresses are still available after these allocations.



Example 6

- **Group 1**

For this group, each customer needs 256 addresses. This means the suffix length is 8 ($2^8 = 256$). The prefix length is then $32 - 8 = 24$.

- The addresses are:

- 1st Customer 190.100.0.0/24 190.100.0.255/24
- 2nd Customer 190.100.1.0/24 190.100.1.255/24
- ...
- 64th Customer 190.100.63.0/24 190.100.63.255/24
- Total = $64 \times 256 = 16,384$



Example 6

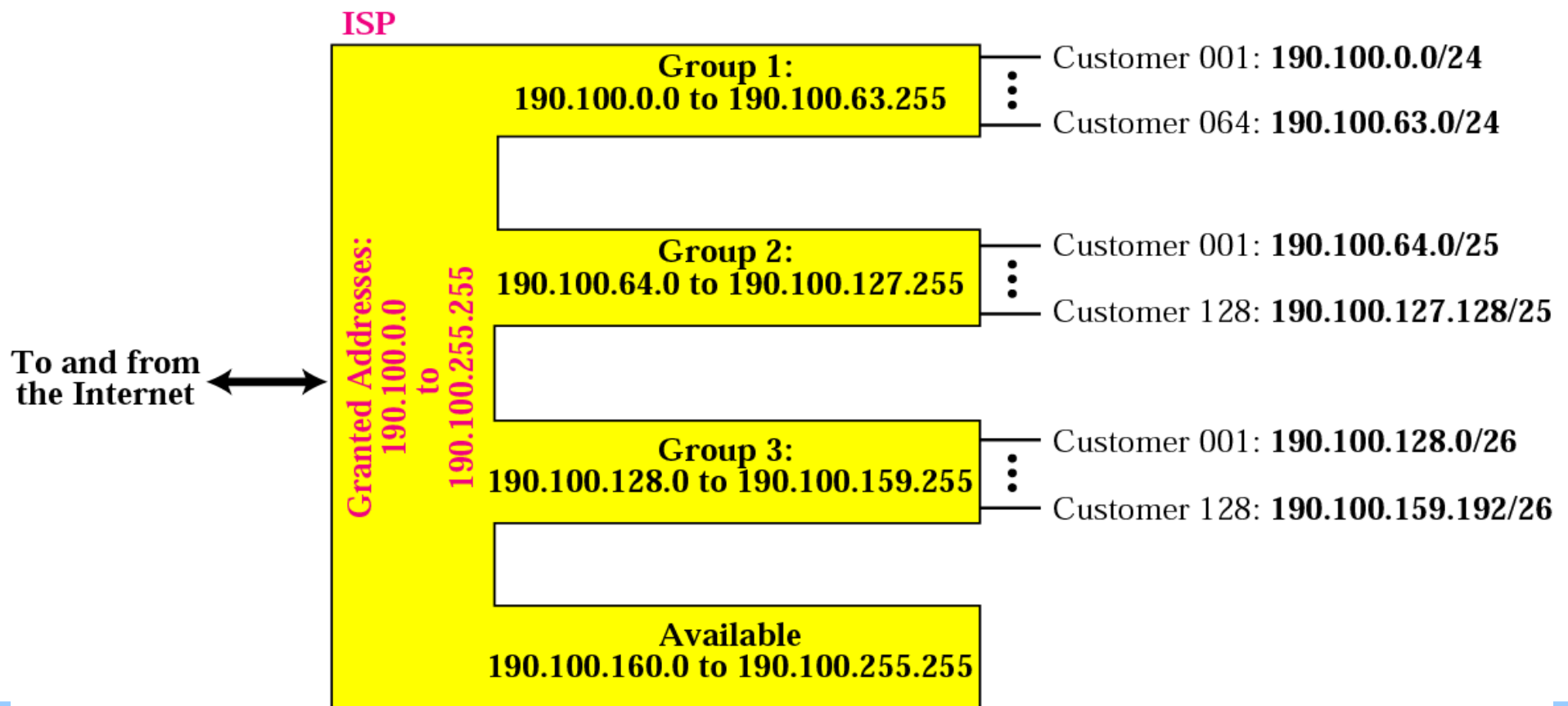
- **Group 3**

For this group, each customer needs 64 addresses. This means the suffix length is 6 ($2^6 = 64$). The prefix length is then $32 - 6 = 26$. The addresses are:

- 1st Customer 190.100.128.0/26 190.100.128.63/26
- 2nd Customer 190.100.128.64/26
190.100.128.127/26
- ...
- 128th Customer 190.100.159.192/26
190.100.159.255/26
- Total = $128 \times 64 = 8,192$

Example 6

- Number of granted addresses to the ISP: 65,536
- Number of allocated addresses by the ISP: 40,960
- Number of available addresses: 24,576





Exercise

- An ISP is granted a block of addresses starting with 50.80.0.0/16.
- The ISP needs to distribute these blocks to 1000 customers as follows:
 - The first group has 200 medium-sized businesses; each needs 128 addresses.
 - The second group has 400 small businesses; each needs 16 addresses.
 - The third group has 2000 households; each needs 4 addresses.
- Design the subblocks and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.



Network Address Translation (NAT)

- NAT enables a user to have a large set of addresses internally and one address, or a small set of addresses, externally
 - The traffic inside can use the large set
 - The traffic outside use the small set.
- To separate the addresses used inside the home or business and the ones used for the Internet, the Internet authorities have reserved three sets of addresses as **private addresses**

<i>Range</i>			<i>Total</i>
10.0.0.0	to	10.255.255.255	2^{24}
172.16.0.0	to	172.31.255.255	2^{20}
192.168.0.0	to	192.168.255.255	2^{16}

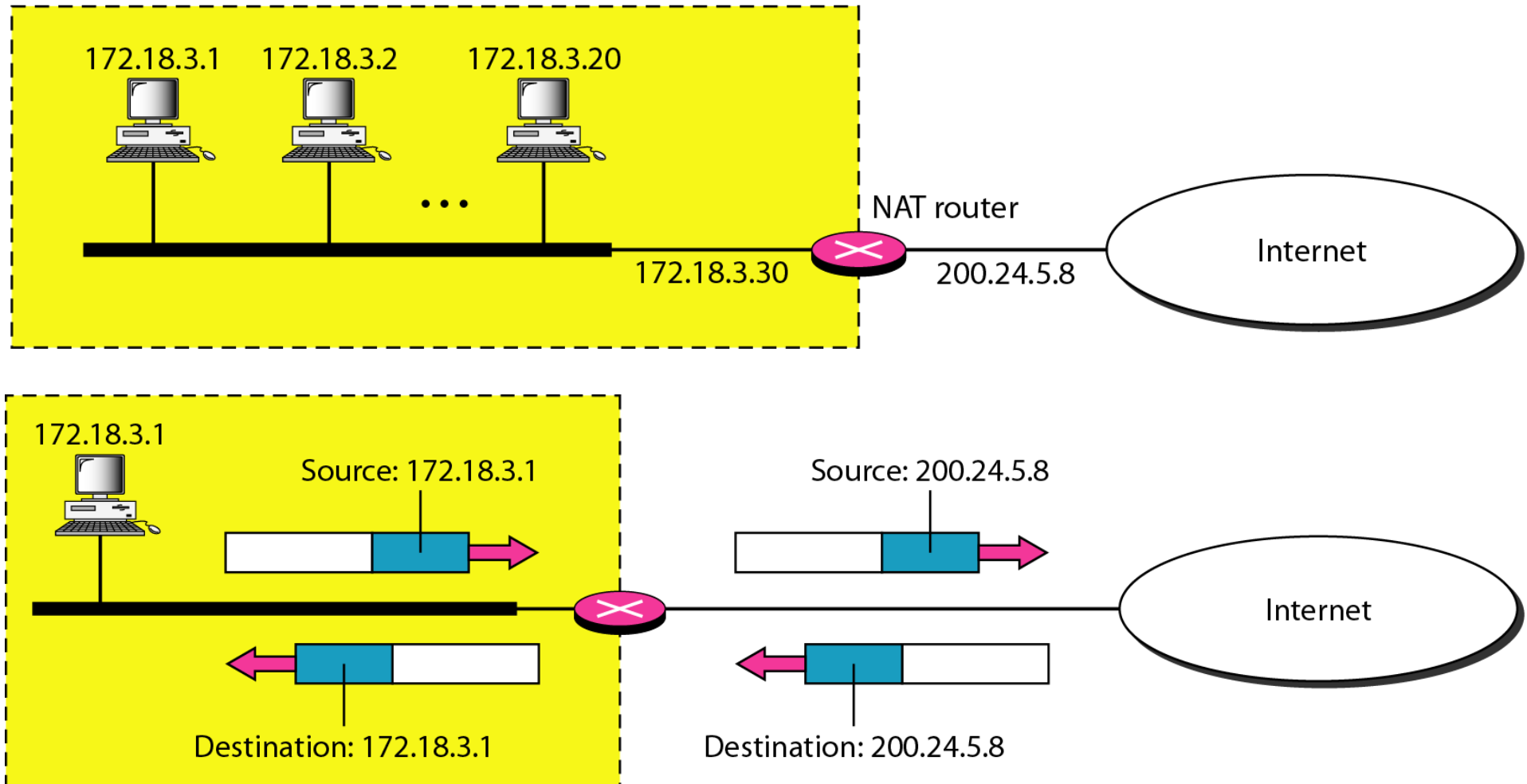


Private addresses

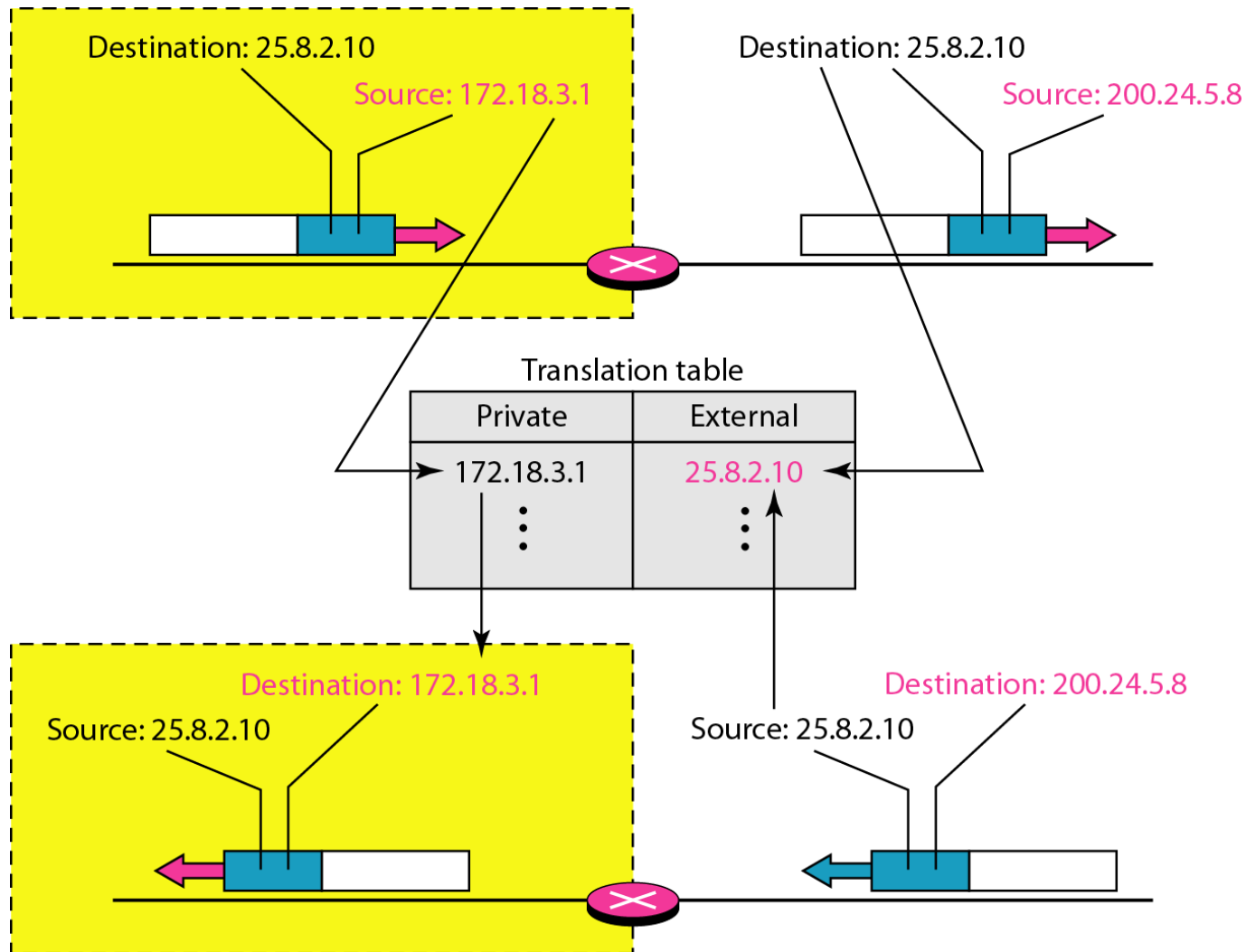
- Any organization can use an address out of this set without permission from the Internet authorities
- They are unique inside the organization, but they are not unique globally
- No router will forward a packet that has one of these addresses as the destination address
- The site must have only one single connection to the global Internet through a router that runs the NAT software

NAT Implementation

Site using private addresses



NAT Address Translation



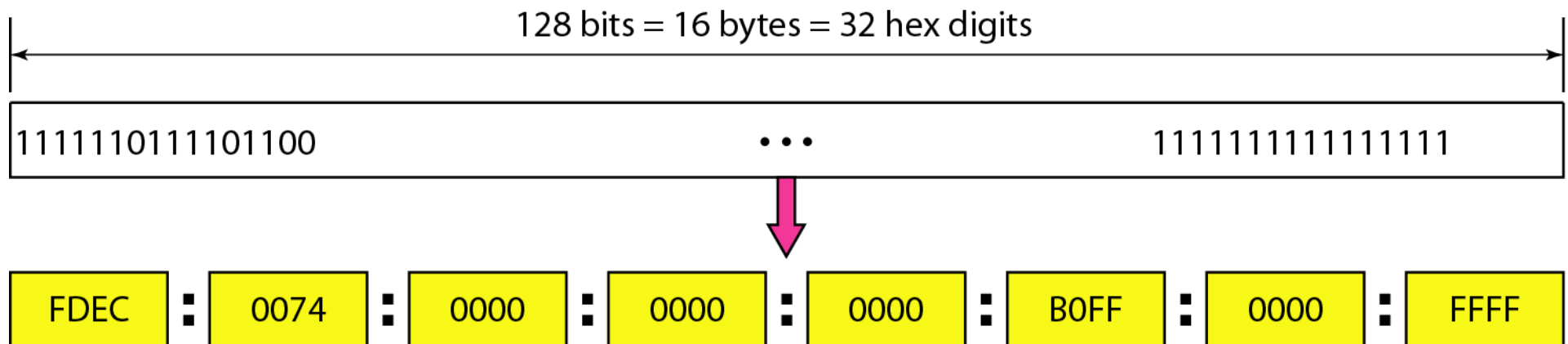


IPv6 Addresses

- Despite all short-term solutions such as classless addressing, DHCP etc. , address depletion is still a long-term problem for the Internet
- In IPv4, lack of accommodation for real-time audio and video transmission, and encryption and authentication of data for some applications, have been the motivation for IPv6

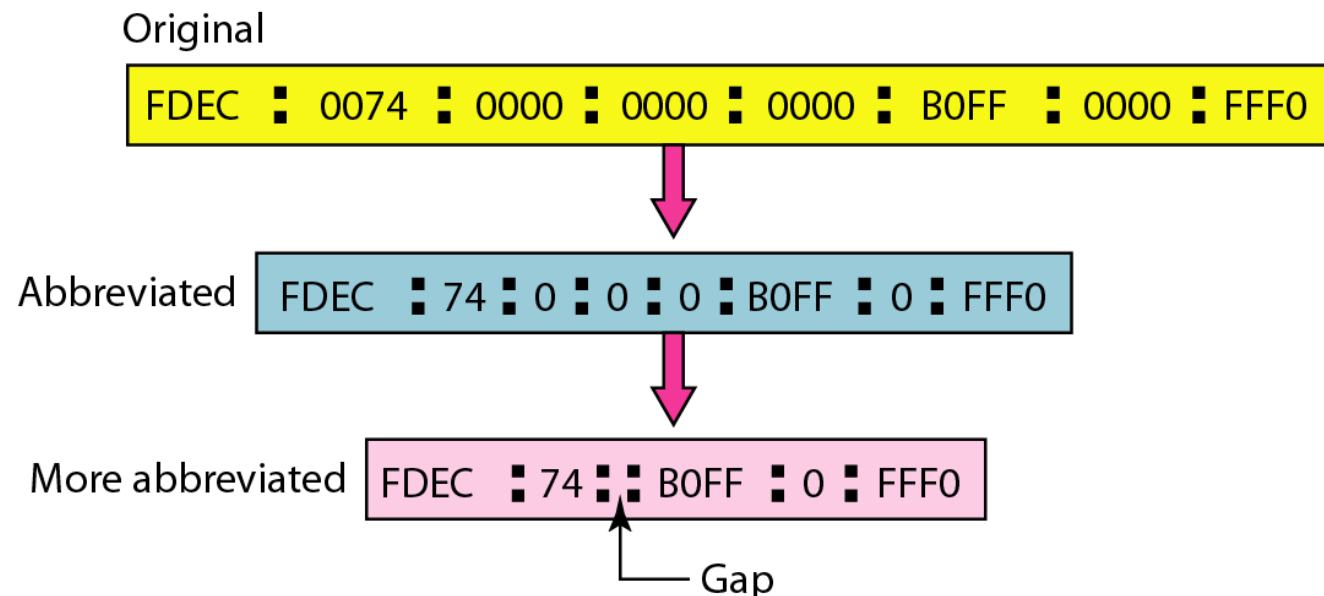
Structure

- An IPv6 address consists of 16 bytes (octets); it is 128 bits long
- IPv6 specifies hexadecimal colon notation



Abbreviaton

- IP address, even in hexadecimal format, is very long, many of the digits are zeros
- In this case, we can abbreviate the address. The leading zeros of a section (four digits between two colons) can be omitted





Address Space

- In IPv6, 2^{128} addresses are available
- A few leftmost bits, called the type prefix, in each address define its category
- The type prefix is variable in length, but it is designed such that no code is identical to the first part of any other code
- IPv6 defines two types of unicast addresses: geographically based and provider-based
- Multicast address in IPv6 used to deliver to a group of IPs
- IPv6 also defines **anycast** addresses, which is like a multicast address. However, a packet destined for an anycast address is delivered to only one of the members of the anycast group, the nearest one
- Reserved addresses start with eight 0s (type prefix is 00000000).



Next : Internet Protocol